9/9/2026

The law of quadratic reciprocity

Thm (Quadratic reciprocity) Let p, q be distinct primes. Then

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}.$$

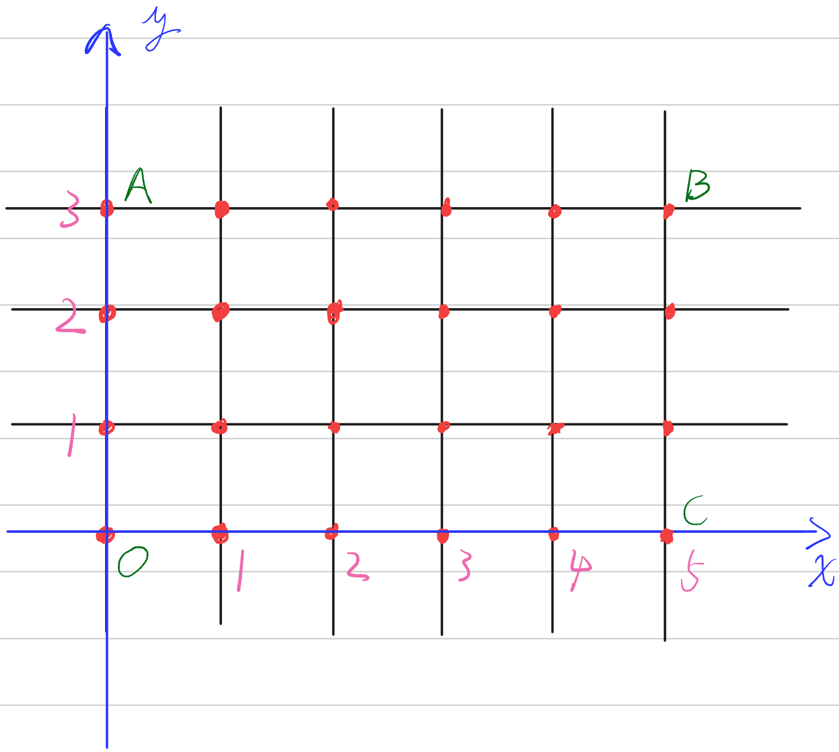
Equivalently,

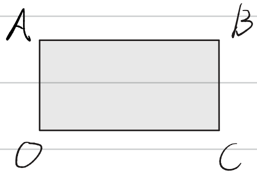
$$\left(\frac{q}{p}\right) = \left(\frac{p}{q}\right),$$

unless $p, q \equiv 3 \pmod 4$, in which case,

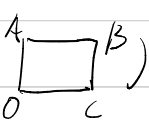
$$\left(\frac{q}{p}\right) = -\left(\frac{p}{q}\right).$$

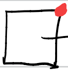
1. Lattice points in a rectangle



Q1: How many lattice points in 

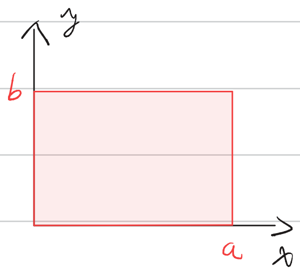
, including points on AB, BC but not those on OC, OA?

A: $15 = 5 \cdot 3 =$ Area(

) : • $\rightleftarrows$  $\to$ Area 1

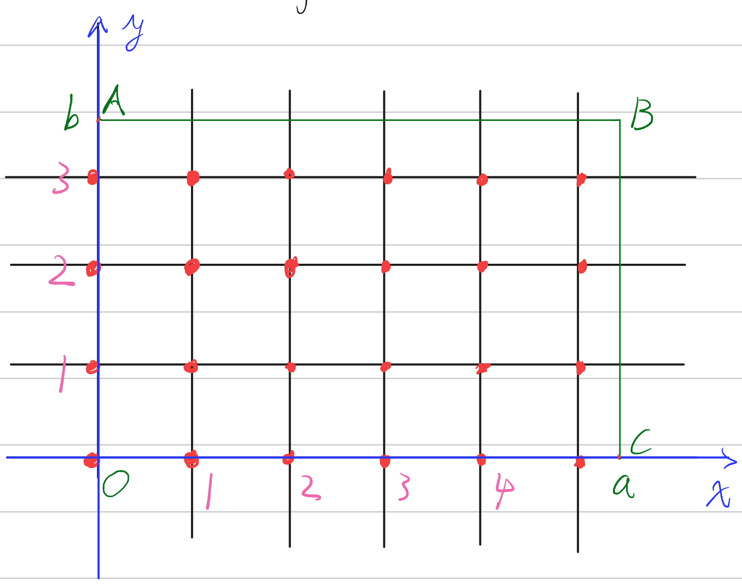
Let a, b be positive integers.

Number of lattice points

in  $= \text{Area} = ab.$

excluding those on the axies

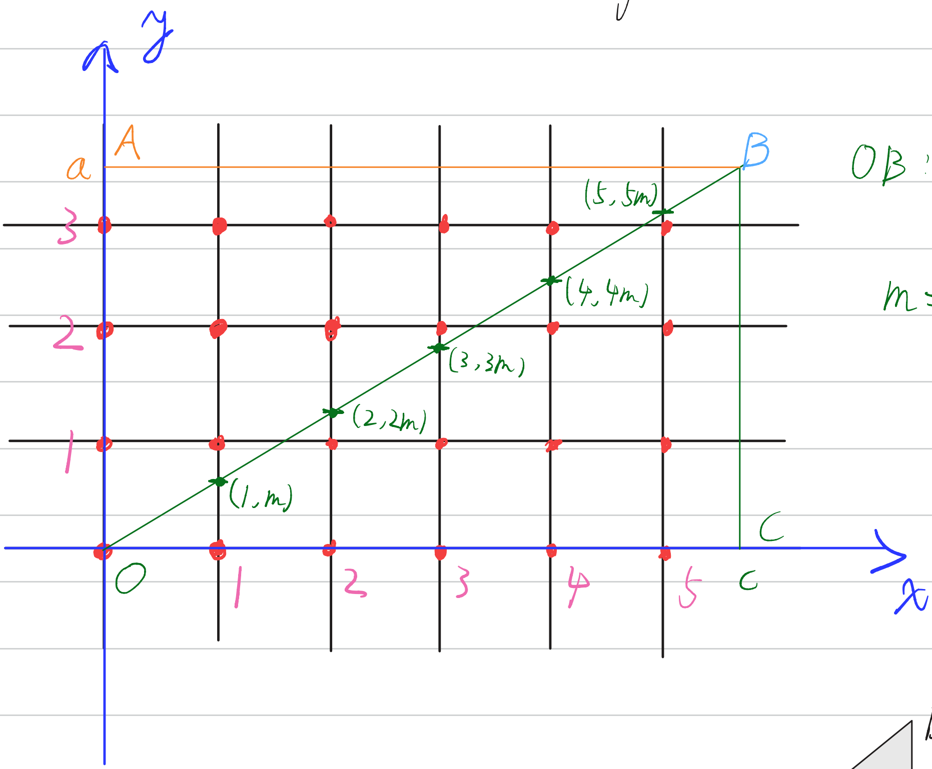
Q2: What if $a, b > 0$ are real numbers?



Number of lattice points

$= \lfloor a \rfloor \cdot \lfloor b \rfloor.$

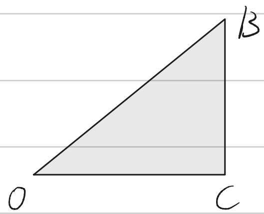
2. Lattice points in a triangle



OB : $y = mx$

$m = \frac{a}{c}$.

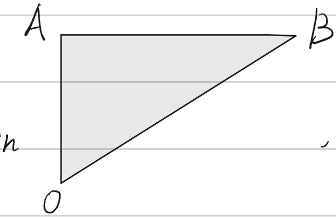
Q1: How many lattice points in



, including points on OB, BC but not those on OC ?

A: $\lfloor 1\cdot m\rfloor + \lfloor 2\cdot m\rfloor + \lfloor 3\cdot m\rfloor + \cdots + \lfloor \lfloor c\rfloor m\rfloor = \sum_{k=1}^{\lfloor c\rfloor} \lfloor \frac{ka}{c} \rfloor$.

Q2: How many lattice points in



, including points on OB, AB but not those on OA?

A: $\lfloor 1\cdot\frac{1}{m}\rfloor + \lfloor 2\cdot\frac{1}{m}\rfloor + \cdots + \lfloor \lfloor a\rfloor\cdot\frac{1}{m}\rfloor = \sum_{k=1}^{\lfloor a\rfloor}\lfloor k\cdot\frac{1}{m}\rfloor = \sum_{k=1}^{\lfloor a\rfloor}\lfloor \frac{kc}{a}\rfloor$

using OB: $x = \frac{1}{m}\cdot y$.

Eisentein's proof of quadratic reciprocity:

By Gauss's lemma,

$$\left(\frac{p}{q}\right) = (-1)^{S(p,q)}, \quad S(p,q) = \sum_{k=1}^{\frac{q-1}{2}} \left\lfloor \frac{kp}{q} \right\rfloor.$$

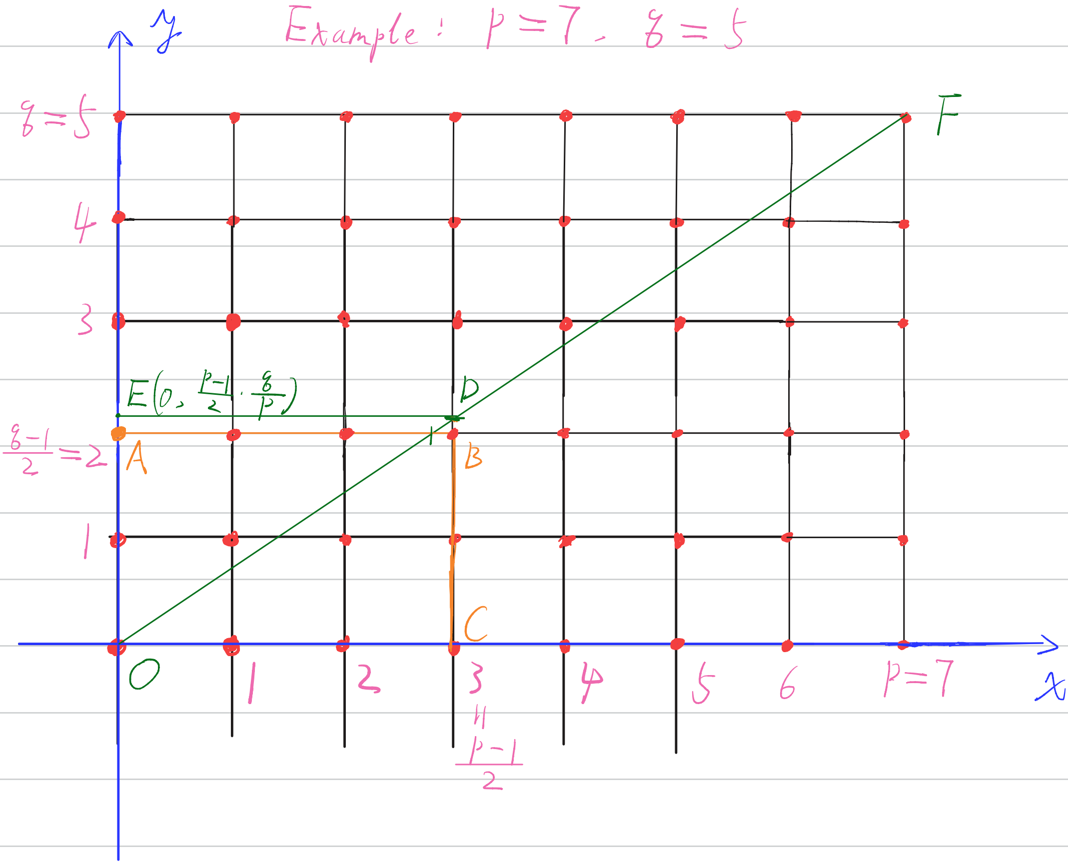
$$\left(\frac{q}{p}\right) = (-1)^{S(q,p)}, \quad S(q,p) = \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{kq}{p} \right\rfloor.$$

$$\Rightarrow \quad \left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{S(p,q)+S(q,p)}$$

$$\overset{?}{=} (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}.$$

Need to show $S(p,q)+S(q,p) = \frac{p-1}{2}\cdot\frac{q-1}{2}$.

Assume $p > q$.



Line OF: $y = \frac{q}{p} \cdot x$.

Claim: There are no lattice points on OD.

For $1 \leqslant x \leqslant \frac{p-1}{2}$, $y = \frac{q}{p} \cdot x \notin \mathbb{Z}$, since $p \neq q \Rightarrow$

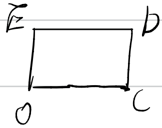
$(p,q)=1$ and $1 \le x < p \Rightarrow p \nmid x$, i.e., $(p,x)=1$.

$$D = \left(\frac{p-1}{2}, \underbrace{\frac{p-1}{2}\cdot\frac{q}{p}}_{\notin \mathbb{Z},\ \text{as } p\nmid q,\ p\nmid(p-1)}\right).$$

Also, since $p>q$, we have

$$\frac{q-1}{2} < \frac{p-1}{2}\cdot\frac{q}{p} < \frac{q+1}{2}.$$

$$\Rightarrow \left\lfloor \frac{p-1}{2}\cdot\frac{q}{p} \right\rfloor = \frac{q-1}{2}.$$

Number of lattice points in 

$= \frac{p-1}{2}\cdot\frac{q-1}{2}$.

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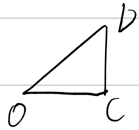
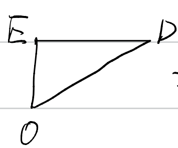
Number of lattice points in 

$= \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{kq}{p} \right\rfloor = S(q,p)$

$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad + \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad +$

Number of lattice points in 

$= \sum_{k=1}^{\frac{q-1}{2}} \left\lfloor \frac{kp}{q} \right\rfloor = S(p,q)$

Eg $\left(\frac{90}{103}\right) = \left(\frac{2\cdot 3^2 \cdot 5}{103}\right) = \left(\frac{2}{103}\right)\underbrace{\boxed{\left(\frac{3}{103}\right)^2}}_{=1}\left(\frac{5}{103}\right)$

prime

$103 = 8\cdot 12 + 7 \equiv 7 \pmod 8 \Rightarrow \left(\frac{2}{103}\right) = 1.$

$5 \equiv 1 \pmod 4$, $103 \equiv 3 \pmod 4$

QR $\Rightarrow \left(\frac{5}{103}\right) = \left(\frac{103}{5}\right) = \left(\frac{3}{\underset{\equiv 1 \pmod 4}{5}}\right) = \left(\frac{5}{3}\right) = \left(\frac{2}{3}\right) = -1.$

So $\left(\frac{90}{103}\right) = -1.$

Jacobi symbol.

Given $a \in \mathbb{Z}$ and a positive odd integer $n = p_1^{m_1} \cdots p_k^{m_k}$,

define

$$\left(\frac{a}{n}\right) = \left(\frac{a}{p_1}\right)^{n_1}\left(\frac{a}{p_2}\right)^{n_2}\cdots\left(\frac{a}{p_k}\right)^{n_k},$$

with $\left(\frac{a}{1}\right) = 1$.

Observation:

$$\left(\frac{a}{n}\right) = \begin{cases} \pm 1, & (a,n)=1, \\ 0, & \text{otherwise} \end{cases}$$

Properties: Let $a, b \in \mathbb{Z}$ and $m, n \geq 1$ be odd integers.

1. (Periodicity) If $a \equiv b \pmod{n}$, then

$$\left(\frac{a}{n}\right) = \left(\frac{b}{n}\right).$$

2. (Complete multiplicativity)

$$\left(\frac{ab}{n}\right)=\left(\frac{a}{n}\right)\left(\frac{b}{n}\right).$$

$$\left(\frac{a}{mn}\right)=\left(\frac{a}{m}\right)\left(\frac{a}{n}\right).$$

(3) (-1 mod n) $\left(\frac{-1}{n}\right)=(-1)^{\frac{n-1}{2}}=\begin{cases}1, & n\equiv 1 \pmod 4\\ -1, & n\equiv 3 \pmod 4\end{cases}$

(4) (2 mod n) $\left(\frac{2}{n}\right)=(-1)^{\frac{n^2-1}{8}}=\begin{cases}1, & n\equiv 1,7 \pmod 8\\ -1, & n\equiv 3,5 \pmod 8\end{cases}$

(5) (Quadratic reciprocity) If $(m,n)=1$, then

$$\left(\frac{m}{n}\right)\left(\frac{n}{m}\right)=(-1)^{\frac{m-1}{2}\cdot\frac{n-1}{2}}$$

$$=\begin{cases}-1, & m,n\equiv 3 \pmod 4\\ 1, & \text{otherwise.}\end{cases}$$

Pf. HW 2.

Warning! If $a \in \mathbb{Z}$ is a quadratic residue n in the sense that $(a, n) = 1$ and there is $x \in \mathbb{Z}$ s.t.

$$x^2 \equiv a \pmod{n},$$

then $\left(\frac{a}{n}\right) = \left(\frac{x^2}{n}\right) = \left(\frac{x}{n}\right)^2 = 1$, as $(a, n) = 1 \Rightarrow (x, n) = 1$.

But the converse is false. For instance,

$$\left(\frac{5}{9}\right) = \left(\frac{5}{3}\right)^2 = 1,$$

but there is no $x \in \mathbb{Z}$ s.t. $x^2 \equiv 5 \pmod{9}$.

Sum of two squares

Q: Which positive integers n can be expressed as

$$n = x^2 + y^2, \quad x, y \in \mathbb{Z} \ ?$$

n	$\square + \square$	n	$\square + \square$
1	$1^2 + 0^2$	13	$2^2 + 3^2$
2	$1^2 + 1^2$	14	X
3	X	15	X
4	$2^2 + 0^2$	16	$4^2 + 0^2$
5	$1^2 + 2^2$	17	$1^2 + 4^2$
6	X	18	$3^2 + 3^2$
7	X	19	X
8	$2^2 + 2^2$	20	$2^2 + 4^2$
9	$3^2 + 0^2$	21	X
10	$1^2 + 3^2$	22	X
11	X	23	X
12	X	24	X